☢ Nuclear Renaissance — Comprehensive Investment Thesis

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1. Executive Summary & Investment Thesis

Why the Nuclear Renaissance Is Real

The convergence of three megatrends has created the most favorable environment for nuclear energy since the Atoms for Peace era:

1. AI Data Center Power Demand — The Insatiable Appetite Big Tech's combined AI infrastructure spend is projected at **$650 billion in 2026** (Amazon, Alphabet, Meta, Microsoft). Each hyperscale data center demands 500MW–2GW of reliable, 24/7 baseload power — exactly what nuclear provides and what renewables alone cannot. Meta alone just signed deals for **6.6 GW of nuclear capacity** across TerraPower, Oklo, and Vistra. Microsoft's 20-year PPA with Constellation to restart Three Mile Island Unit 1 (now "Crane Clean Energy Center") signals that Big Tech is willing to underwrite nuclear economics directly.

2. Policy Tailwinds — Executive Order 14300 & NRC Reform President Trump's EO 14300 mandates comprehensive NRC reform with **fixed licensing deadlines: 18 months maximum for new-reactor construction-and-operation applications, 12 months for license renewals**. The NRC has launched a major reorganization to comply, including new offices and streamlined review processes. A proposed microreactor licensing rule is targeted for March 2026, with a final rule by September 2026. The ADVANCE Act (bipartisan) further accelerates advanced reactor deployment. This is the most pro-nuclear regulatory environment in decades.

3. Energy Security & Grid Reliability Russian enrichment dependence (~40% of global SWU capacity) has made domestic nuclear fuel a national security imperative. Grid reliability concerns — from Texas winter storms to California rolling blackouts — have rehabilitated nuclear's reputation as the ultimate baseload source. The conversation has shifted from "how do we shut them down?" to "how fast can we build them?"

The Investment Thesis in One Sentence

Nuclear is the only energy source that satisfies Big Tech's simultaneous demands for massive baseload capacity, 24/7 reliability, and carbon-free credentials — and the policy environment has never been more supportive.

The Catch

Much of this is already priced in. Nuclear stocks have run 50–200%+ in the past year. The thesis is about separating the **real operators generating cash today** from the **speculative plays that won't generate a single watt for 5–10 years**. As someone with 43 years of NRC licensing experience, Curtis, you have a unique edge in evaluating who's real and who's running a PowerPoint reactor.

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2. Overall Rankings Table

Rank	Ticker	Price	Mkt Cap	P/E	Fwd P/E	Rev (TTM)	FCF (TTM)	Beta	Risk	Tier	Key Strength
1	**CEG**	$288	$105B	33.0	25.1	$24.8B	$3.1B	1.14	Med (4)	Core	Largest US nuclear fleet (21 reactors), TMI restart + Microsoft PPA
2	**VST**	$171	$58B	61.7	18.7	$17.2B	$1.8B	1.44	Med (4)	Core	Comanche Peak operator, Meta deal, 5GW data center campus nearby
3	**GEV**	$802	$218B	45.2	35.6	$38.1B	$5.3B	N/A	Low (2)	Core	Diversified energy tech, BWRX-300 SMR, turbine/generator services
4	**CCJ**	$113	$49B	114.0	59.7	$3.5B	$348M	N/A	Med (5)	Core	Uranium supply dominance + Westinghouse ownership
5	**BWXT**	$200	$18B	59.8	46.6	$3.1B	$273M	0.83	Med (4)	Core	Naval nuclear monopoly, TRISO fuel, defense moat
6	**LEU**	$199	$3.9B	51.1	37.9	$449M	$38M	1.21	High (10)	Speculative	Only US enrichment, HALEU critical for advanced reactors
7	**FLR**	$45	$7.4B	2.3	19.5	$15.6B	$299M	1.38	High (9)	Watchlist	Nuclear construction expertise, NuScale investor
8	**OKLO**	$66	$10.3B	N/A	N/A	$0	-$41M	0.77	Med (5)	Speculative	Sam Altman-backed, Meta deal, 91mo cash runway
9	**SMR**	$14	$4.0B	N/A	N/A	$64M	$298M*	2.16	High (8)	Speculative	Only NRC-approved SMR design, Fluor spinoff
10	**NNE**	$25	$1.3B	N/A	N/A	$0	-$25M	N/A	High (7)	Watchlist	Micro-reactor concept, 33mo cash runway
11	**URA**	$52	ETF	—	—	—	—	—	Med	Reference	Global X Uranium ETF — broad sector exposure

SMR FCF appears inflated by one-time items; operating cash flow is deeply negative.

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3. Tier Breakdown

Tier 1 — Core Holdings (Proven Operators & Supply Chain)

CEG — Constellation Energy ($288 | P/E 33 | Fwd P/E 25.1)

The Nuclear Fleet King

Constellation operates **21 nuclear reactors** — the largest commercial nuclear fleet in the United States. This isn't optionality; this is **21 operating licenses, 21 reactor vessels producing power and cash flow right now**.

- **Crane Clean Energy Center (TMI restart):** 20-year PPA with Microsoft, $1.6B investment. If successful, this single project proves the "reactor restart" thesis and opens the door for every shuttered plant in America.
- **Meta deal:** Supports relicensing/extended operations of Constellation's Illinois nuclear fleet.
- **Revenue:** $24.8B TTM, $3.1B FCF — this is a real business with real cash.
- **Pulled back 21% YTD** from highs — better entry point than 3 months ago.
- **Risk:** TMI restart execution, regulatory approval timeline, elevated P/E for a utility.
- **Why it's #1:** When Big Tech needs 1GW of clean baseload power *yesterday*, they call CEG because CEG has reactors *running today*.

VST — Vistra Corp ($171 | P/E 61.7 | Fwd P/E 18.7)

Curtis's Home Turf — Deep Dive in Section 5

Vistra operates **Comanche Peak Nuclear Power Plant** (Units 1 & 2, ~2.4 GW combined) in Somervell County, Texas — where Curtis spent 43 years. But the explosive development is the **Comanche Circle project**: nine data center campuses with up to **5 GW of combined capacity** proposed near Comanche Peak. Meta has signed a deal with Vistra for power from Ohio and Pennsylvania plants.

- **EBITDA Guidance:** $6.8B–$7.6B for FY2026 — a transformational step.
- **Forward P/E of 18.7** is the cheapest in the nuclear group on a forward basis.
- **Strong Buy consensus** from analysts.
- **Risk:** Texas grid volatility, Moss Landing battery fire cleanup (late 2026), execution on data center partnerships.

GEV — GE Vernova ($802 | P/E 45.2 | Fwd P/E 35.6)

The Picks-and-Shovels Play

GE Vernova is the energy technology spinoff from GE — turbines, generators, grid equipment, and the **BWRX-300 small modular reactor** design. Every nuclear plant in America needs turbine/generator services. Every grid expansion for data centers needs GEV's equipment.

- **$38.1B revenue, $5.3B FCF** — the largest and most diversified player in this portfolio.
- **BWRX-300:** A boiling water SMR design currently in NRC review. Based on proven BWR technology (GE designed the BWR fleet). Lower risk than molten salt or sodium-cooled designs because the physics are well understood.
- **Risk:** Lowest risk rating (2) in the group. Premium valuation but justified by diversification.
- **Why it's here:** You don't need to pick the winning reactor design. GEV wins regardless — they service the entire installed fleet AND have their own SMR optionality.

CCJ — Cameco ($113 | P/E 114 | Fwd P/E 59.7)

Uranium Supply Chain + Westinghouse

Cameco is the world's largest publicly traded uranium producer AND owns Westinghouse Electric (the company that designed Comanche Peak's reactors, Curtis). This is the ultimate picks-and-shovels play.

- **Uranium at $101/lb** — back in triple digits with a 24% surge in January 2026. Structural supply deficit with demand projected to rise 28% by decade's end and double by 2040.
- **Westinghouse:** Designs, fuels, and services ~50% of the world's operating reactors. AP1000 is the most recent NRC-certified large reactor design.
- **Risk:** Uranium price volatility, Kazatomprom supply decisions, elevated P/E (pricing in sustained uranium bull market).

BWXT — BWX Technologies ($200 | P/E 59.8 | Fwd P/E 46.6)

The Naval Nuclear Monopoly

BWXT is the **sole manufacturer of nuclear reactors for the U.S. Navy** — every submarine and aircraft carrier reactor goes through BWXT. They also produce TRISO fuel (the "meltdown-proof" fuel used in advanced reactors like Kairos Power's Hermes).

- **28.9% revenue growth** — defense spending is flowing.
- **TRISO fuel production** positions them as essential infrastructure for the advanced reactor ecosystem. Whether Kairos, X-energy, or anyone else succeeds, they likely need BWXT's fuel.
- **Risk:** Sole-source dependency on Navy contracts, high P/E, limited commercial exposure.
- **Moat:** You can't build a new BWXT. The nuclear-grade manufacturing capabilities, ASME Section III certifications, and security clearances took decades to establish.

Tier 2 — Speculative Positions (High Risk, High Optionality)

LEU — Centrus Energy ($199 | P/E 51.1)

Only US Uranium Enrichment

Centrus operates the **only active uranium enrichment facility in the United States** (Piketon, Ohio) and is the only US company producing HALEU — High-Assay Low-Enriched Uranium — required by virtually every advanced reactor design (OKLO, Kairos, TerraPower, X-energy).

- **Strategic irreplaceability:** With ~40% of global enrichment capacity in Russia (TENEX/Rosatom), US HALEU supply is a national security imperative. LEU is the only game in town.
- **Short interest: 25%** — the bears are aggressive. This cuts both ways.
- **Risk:** Execution on HALEU scale-up, DOE contract dependency, volatile trading due to small float + high short interest.
- **Position sizing:** Keep small (5%). If HALEU demand materializes with DOE production contracts, this becomes a much bigger position.

OKLO ($66 | No Revenue | 91mo Cash Runway)

Sam Altman's Nuclear Bet — Deep Dive in Section 6

Oklo is developing the Aurora fast reactor — a metallic fuel, sodium-cooled fast reactor. Sam Altman (OpenAI CEO) is chairman. The NRC **denied Oklo's first combined license application in 2022** for insufficient safety analysis, but they've resubmitted with substantially more data.

- **Meta deal:** 1.2 GW plant in Ohio, potentially online by 2030. Plus DOE Reactor Pilot Program selection.
- **Fuel recycling:** DOE grant for spent fuel recycling R&D — converts nuclear waste into Aurora fuel. If it works, this is transformational for the entire industry.

- **CEO selling:** Jacob DeWitte sold $15M in stock in early February 2026. Not ideal.
- **Risk:** NRC approval uncertain (denied once), no revenue, no operating reactor, insider selling.
- **Bull case:** If Aurora gets licensed and the fuel recycling works, OKLO becomes the most vertically integrated nuclear company in America.

SMR — NuScale Power ($14 | Pre-Revenue | Beta 2.16)

Only NRC-Approved SMR Design — Deep Dive in Section 6

NuScale has the **only NRC-approved small modular reactor design in the United States** — a 77 MWe light-water SMR using natural circulation (no coolant pumps — a genuine safety advantage Curtis would appreciate). But the flagship Idaho project was cancelled in 2023 when costs soared and utility partners backed out.

- **Design cert is real** — NRC Standard Design Approval (2023) is a genuine technical achievement. No one else has this.
- **Fluor spinoff:** Fluor Corp is divesting its stake, which creates both uncertainty and a potential catalyst.
- **Short interest: 23.5%** — market is deeply skeptical.
- **Price collapsed:** From $28 to $14 — down 50% in recent months. The market is punishing the lack of a flagship project.
- **Risk:** No signed construction agreement, negative operating margins, high short interest, Fluor exit uncertainty.

Tier 3 — Watchlist (Don't Buy Yet)

NNE — Nano Nuclear Energy ($25 | No Revenue | 33mo Cash Runway)

- **Concept-stage** micro-reactor company. No NRC application filed. No prototype built.
- **33 months of cash runway** — shortest in the group. Dilution risk is high.
- **30% short interest** — market says this is hype, not hardware.
- **Revisit in 12–18 months** if they file an NRC application and demonstrate prototype progress.

FLR — Fluor Corp ($45 | P/E 2.3 | Fwd P/E 19.5)

- Nuclear construction and decommissioning expertise is valuable, but Fluor has **terrible cash flow quality** (OCF/NI = 0.09 historically).
- Spinning off NuScale stake — removes nuclear optionality.
- Revenue declining 18% YoY.
- **Wait for the business to stabilize** before committing capital.

URA — Global X Uranium ETF ($52)

- Broad uranium sector exposure including miners, enrichers, and producers globally.
- **Use as a reference** for uranium market sentiment.
- If you want diversified uranium exposure without single-stock risk, URA is the vehicle — but you already have CCJ and LEU for more concentrated bets.

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4. Recommended $100K Allocation

Strategy: Barbell — Heavy on Proven Operators, Small Speculative Positions

Ticker	Tier	Allocation	Amount	~Shares	Rationale
CEG	Core	25%	$25,000	87	Anchor. Largest nuclear fleet, Microsoft PPA, 21% pullback = entry point
VST	Core	20%	$20,000	117	Domain expertise edge. Comanche Peak + data center campus. Cheapest Fwd P/E (18.7)
GEV	Core	12%	$12,000	15	Diversified energy tech, BWRX-300, services every nuclear plant
CCJ	Core	12%	$12,000	106	Uranium supply + Westinghouse. Picks-and-shovels
BWXT	Core	10%	$10,000	50	Naval nuclear monopoly, TRISO fuel, defense moat
LEU	Speculative	5%	$5,000	25	Only US enrichment. HALEU strategic asset. High risk
OKLO	Speculative	5%	$5,000	76	Meta deal, DOE pilot, Sam Altman backing. Watch NRC closely
SMR	Speculative	3%	$3,000	210	Only NRC-approved SMR design. Beaten down, potential turnaround
URA	Reference	3%	$3,000	58	Broad uranium ETF for sector exposure
Cash	Reserve	5%	$5,000	—	Dry powder for pullbacks or adding to winners
		100%	**$100,000**		

Portfolio Characteristics

- **Core Holdings:** 79% in profitable, cash-generating companies
- **Speculative:** 13% in pre-revenue/high-risk names
- **Cash Reserve:** 5% for opportunistic adds
- **Effective Sector Beta:** ~1.1 (moderate cyclicality)
- **Dividend Yield:** ~1.5% weighted average (CEG, VST, CCJ pay dividends)

Entry Strategy

1. **Scale in over 4–6 weeks.** Nuclear names are volatile. Don't buy all at once.
2. **CEG and VST** have pulled back significantly — best near-term entry points of the group.
3. **CCJ at $113** is near highs — consider waiting for a pullback to $100–105 range.
4. **GEV at $802** — up significantly from spinoff. Leg in with half-position, add on any 10%+ correction.
5. **SMR at $14** is near 52-week lows — if you're going to take the speculative bet, this is a better entry than $28.
6. **Never chase** any name that's up >5% on a given day.

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5. Deep Dive: VST/Vistra — Curtis's Domain Edge

Comanche Peak Nuclear Power Plant

Curtis, you know this plant better than any analyst on Wall Street. Here's what makes it interesting from an investment perspective:

Plant Specs: - **Unit 1:** Westinghouse 4-loop PWR, ~1,200 MWe, COL 1990 - **Unit 2:** Westinghouse 4-loop PWR, ~1,200 MWe, COL 1993 - **Combined capacity:** ~2,400 MWe - **License expiration:** Unit 1 (2030), Unit 2 (2033) — **license renewal is a near-term catalyst** - **Capacity factor:** Consistently >90% — among the best-operated plants in the fleet - **Location advantage:**

ERCOT (Texas deregulated market) — power prices are higher and more volatile than regulated markets, meaning higher revenue per MWh

The Comanche Circle Data Center Campus — Game Changer

This is the development that transforms VST from "utility with nuclear" to "AI infrastructure play":

- **Nine data center campuses** proposed near Comanche Peak
- **Up to 5 GW of combined capacity** — that's more than double Comanche Peak's output
- **Somervell County** rejected a development moratorium — the community is welcoming it
- **Proximity to Comanche Peak** = direct grid access to nuclear baseload power
- **ERCOT market** = premium power pricing, especially during peak demand

Why Curtis's Experience Is the Edge

As someone who worked I&C systems, QA, and licensing at Comanche Peak for 43 years, you can evaluate:

1. **License renewal readiness:** You know the plant's aging management programs, whether the concrete is sound, whether the cable qualification is adequate, whether the reactor vessel embrittlement is within limits. Wall Street analysts read the 10-K. You've walked the containment.

2. **Uprate potential:** Has Comanche Peak explored measurement uncertainty recapture (MUR) uprates or extended power uprates (EPU)? Each percentage point of uprate is additional revenue at minimal marginal cost.

3. **Operational culture:** You know whether the plant's safety culture is genuine or performative. NRC inspection findings tell you something; 43 years inside tells you everything.

4. **Data center proximity risks:** You understand the grid interconnection challenges — can the existing switchyard and transmission lines handle 5 GW of additional load? What are the voltage regulation implications?

5. **Spent fuel storage:** You know the dry cask storage situation at Comanche Peak. Any pending ISFSI expansion? This matters for continued operation timelines.

Vistra Financial Snapshot

- **EBITDA guidance: $6.8B–$7.6B** (FY2026) — transformational
- **Forward P/E: 18.7** — cheapest in the nuclear group
- **Diversified generation:** Nuclear, natural gas, solar, battery storage (not just nuclear)
- **Moss Landing risk:** Battery fire cleanup ongoing, full restoration not until late 2026
- **Meta partnership:** Power sourcing from Vistra's Ohio and Pennsylvania plants

VST Investment Verdict

Strong Buy for this portfolio. The combination of proven nuclear operations, the Comanche Circle data center development, Meta partnership, and Curtis's unparalleled domain expertise makes this the highest-conviction position in the portfolio. The forward P/E of 18.7 suggests the market hasn't fully priced in the AI data center catalyst.

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6. Deep Dive: The SMR Race — SMR vs OKLO & the Molten Salt Landscape

The Licensing Timeline = The Key Differentiator

Curtis's thesis — and it's a good one — is that **NRC licensing timeline is the single most important variable** for which advanced reactor company wins AI data center contracts. Big Tech doesn't care about theoretical reactor superiority. They care about: *When can you deliver megawatts to my data center?*

Here's the licensing landscape as of February 2026:

NuScale (SMR) — The Regulatory Leader

Milestone	Status
NRC Design Certification	✅ **Approved** (Standard Design Approval, 2023) — **Only company with this**
Construction Permit	❌ None (Idaho project cancelled 2023)
Operating License	❌ None
Flagship Project	❌ Cancelled — seeking new site/customer
Technology	Light-water PWR, natural circulation, 77 MWe modules
Estimated First Operation	2031–2033 (if a new project is signed soon)

Assessment: NuScale has the regulatory crown jewel — actual NRC design certification — but no customer. The design uses proven light-water technology with no coolant pumps (natural circulation), which is a genuine safety innovation Curtis would recognize. But design cert ≠ operating reactor. They need a site, a utility partner, and a construction/operating license. The clock hasn't started on the hard part.

Oklo — The Silicon Valley Darling

Milestone	Status
NRC Combined License Application	❌ **Denied (2022)** — insufficient safety analysis. Resubmitted.
DOE Pilot Program	✅ Selected for Reactor Pilot Program at INL
Technology	Metallic fuel, sodium-cooled fast reactor (Aurora), 15 MWe
HALEU Fuel	Building fuel fabrication facility at INL
Key Partnerships	Meta (1.2 GW Ohio plant), DOE fuel recycling grant
Estimated First Operation	2028–2030 (DOE pilot), commercial 2030+

Assessment: Oklo is pursuing the most ambitious path — combined license (design + construction + operating in one application), fast reactor technology, fuel recycling. The NRC denial was a serious setback. Their resubmission needs to address the deficiencies. The DOE Reactor Pilot Program is a parallel path that bypasses NRC (DOE self-regulates its own sites), which could get a demonstration reactor operating sooner. But the 1.2 GW Ohio commercial plant for Meta? That's NRC jurisdiction, and Oklo has never successfully navigated it.

The Molten Salt Reactor (MSR) Landscape — Private Companies to Watch

These aren't publicly traded, but they're the competitive context:

Natura Resources — Leading the MSR NRC Race

- **100 MW molten salt reactor** design
- **January 2026:** DOE allocated historic FLiBe (lithium-7 fluoride) salt from the original MSRE at Oak Ridge — this is a material advantage no one else has
- **NGL Water Solutions partnership:** Combining MSR power with Permian Basin produced water treatment — brilliant use case (desalination is energy-intensive)
- **NRC Status:** Leading among MSR developers in licensing engagement
- **Curtis's Take:** Natura has the most pragmatic approach — they're not trying to change the world, they're trying to solve a specific industrial problem (produced water) with proven physics

Kairos Power — The TRISO-Salt Hybrid

- **Hermes test reactor** in Oak Ridge, TN — **NRC construction permit approved** (first non-water-cooled reactor approved in 50 years)
- **14 NRC topical reports approved** as of January 2026 — building the licensing case methodically
- **Technology:** TRISO fuel particles in fluoride salt coolant (FLiBe) — not a true MSR (fuel is solid TRISO, not dissolved in salt), but gets many MSR benefits (low pressure, high thermal efficiency)
- **Google partnership** for power supply
- **BWXT makes their TRISO fuel** — another reason to own BWXT
- **Estimated first criticality:** Late 2026–2027 for Hermes test reactor

Terrestrial Energy — IMSR (Integral Molten Salt Reactor)

- **Canadian company** with NRC pre-application engagement
- **Mid-stage licensing** — further behind Natura and Kairos
- **IMSR design:** True molten salt reactor with fuel dissolved in fluoride salt
- **Timeline:** Several years behind the leaders

TerraPower — The Gates-Backed Natrium

- **Not actually MSR** — Natrium is a sodium-cooled fast reactor with molten salt energy storage
- **Meta deal:** Up to 8 reactors, 1.2 GW, online potentially by 2030
- **Kemmerer, WY:** Construction underway on first commercial Natrium reactor
- **NRC accelerated review:** EO 14300 + DOE support
- **Private company** (Bill Gates-backed) — not directly investable

ACU MSRR — The Research Reactor

- **Abilene Christian University's Molten Salt Research Reactor**
- **NRC construction permit under review** — research reactor, not commercial
- **Expected deployment:** Later 2026
- **Significance:** First university MSR, validates molten salt licensing pathway

Curtis's Licensing Timeline Framework

From 43 years of watching NRC processes, here's the reality:

Company	NRC Status	Years to First Watt	AI Data Center Contract Risk
CEG (existing fleet)	Licensed & operating	**0 — generating NOW**	Lowest
VST (Comanche Peak)	Licensed & operating	**0 — generating NOW**	Lowest
Kairos Power	Construction permit ✅	3–5	Medium
NuScale (SMR)	Design cert ✅	5–8	High (no project)
TerraPower	Accelerated review	4–6	Medium-High
Natura Resources	Pre-application	5–8	High
Oklo	Denied, resubmitted	5–10	Very High
NNE	No application	8–12+	Extreme

The bottom line: Big Tech is signing MOUs with everyone, but the megawatts flowing to data centers in 2026–2030 will come from **existing nuclear plants** (CEG, VST) and possibly **Kairos/TerraPower** if construction stays on schedule. SMR and OKLO are 2030+ stories at best.

SMR vs OKLO — Head to Head

Factor	SMR (NuScale)	OKLO
NRC Design Approval	✅ Yes	❌ Denied, resubmitted
Technology Risk	Lower (light-water, proven physics)	Higher (fast reactor, metallic fuel)
Customer Contracts	❌ None signed post-Idaho cancellation	✅ Meta 1.2 GW
Revenue	$64M (licensing fees)	$0
Cash Runway	Fluor-backed, ~adequate	91 months
Market Cap	$4.0B	$10.3B
Short Interest	23.5%	13.8%
CEO Insider Selling	CFO selling	CEO selling ($15M in Feb)
Price Action	$14 (near 52-wk low)	$66 (down from $194 high)
Valuation Sanity	More reasonable at $14	$10B for zero revenue is aggressive

Curtis's edge here: You understand that NRC design certification is an enormous achievement that took NuScale years. You also know that a denied license application (OKLO) means the NRC found fundamental deficiencies in the safety case. The resubmission needs to address every single deficiency finding — that's not a quick fix. Wall Street treats these as equivalent "nuclear startups." They are not.

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7. Key Risks

Regulatory Risk

Risk	Probability	Impact	Detail
EO 14300 implementation delays	Medium	High	18-month licensing mandate is ambitious. NRC staff may resist or lack capacity.
NRC reorganization disruption	Medium	Medium	Restructuring during heavy licensing workload could slow everything down.
10 CFR 53 delays	Medium	High	New Part 53 framework for advanced reactors still being finalized. Regulatory uncertainty adds years.
Political reversal (2028+)	Low-Medium	Severe	If administration changes, pro-nuclear EOs could be reversed.
Safety event anywhere globally	Low	Catastrophic	A single nuclear incident (even minor) could freeze all licensing and public sentiment overnight.

Valuation Risk

Risk	Probability	Impact	Detail
AI data center narrative fades	Medium	Severe	If AI buildout slows or PPAs don't materialize, the nuclear premium evaporates.
Sector rotation	Medium	Medium	Nuclear names have already run 50–200%+. Momentum can reverse quickly.
Interest rate sensitivity	Medium	Medium	Nuclear projects are capital-intensive. Higher-for-longer rates increase financing costs for new builds.
Speculative name blowups	High	Medium	SMR, OKLO, NNE could easily lose 50%+ on any negative news. Position sizing is critical.

Uranium Supply Risk

Risk	Probability	Impact	Detail
Uranium price correction	Medium	Medium	$101/lb spot has priced in significant demand growth. Kazatomprom supply decisions could swing prices.
Russian enrichment disruption	Low-Medium	High	Sanctions escalation or supply weaponization would spike enrichment costs. Bullish for LEU.
HALEU supply bottleneck	High	High	Every advanced reactor needs HALEU, and only LEU/Centrus can produce it domestically. Bottleneck = delays.

Execution Risk

Risk	Probability	Impact	Detail
Vogtle syndrome	Medium	Severe	Plant Vogtle went from $14B to $35B+ and 7 years late. This ghost haunts every new-build.
Supply chain constraints	High	Medium	Nuclear-grade components (ASME Section III) have limited suppliers. BWXT is one of few. Bottleneck = delays.
Workforce shortage	High	Medium	Qualified nuclear welders, NDE techs, I&C engineers are scarce. The industry lost a generation post-Fukushima.
TMI restart failure	Low-Medium	Severe	If CEG's TMI restart stalls, it delegitimizes the entire "reactor restart" narrative.

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8. Catalysts & Timeline

2026 Catalysts (Near-Term)

Date	Event	Tickers Affected
March 2026	NRC proposed rule for microreactor licensing	NNE, OKLO
Q1 2026	Vistra FY2025 earnings ($6.8B–$7.6B EBITDA guidance)	VST
Q1 2026	CEG earnings + TMI restart progress update	CEG
Mid-2026	NRC reorganization completion per EO 14300	All
Sept 2026	NRC final microreactor licensing rule	NNE, OKLO
Late 2026	Kairos Power Hermes test reactor — potential first criticality	BWXT (TRISO fuel)
Late 2026	ACU MSRR deployment	MSR sector sentiment
2026	Fluor/NuScale spinoff completion	SMR, FLR

2027–2028 Catalysts (Medium-Term)

Date	Event	Tickers Affected
2027	Kairos Hermes operating data available	GEV, BWXT
2027	TMI Unit 1 restart target (Crane Clean Energy Center)	CEG
2027–2028	Comanche Peak license renewal applications	VST
2027–2028	TerraPower Natrium construction milestones (Kemmerer, WY)	Sector sentiment
2028	OKLO Aurora DOE pilot reactor (if on schedule)	OKLO

2029–2030+ Catalysts (Long-Term)

Date	Event	Tickers Affected
2029–2030	NuScale first commercial reactor (if project signed)	SMR
2030	Meta nuclear deals begin delivering power (TerraPower, OKLO)	OKLO, VST
2030+	Uranium demand rise 28% by decade end	CCJ, LEU, URA
2030+	New reactor construction (if NRC licensing accelerates)	All

Macro / Policy Catalysts (Ongoing)

- **ADVANCE Act implementation** — bipartisan nuclear support
- **DOE HALEU production scale-up** — critical for LEU
- **Uranium spot price** — $101/lb and climbing; structural bull market
- **Big Tech AI capex** — $650B projected for 2026; nuclear PPAs follow the money
- **UNECE regulatory harmonization** — could open international markets for US reactor designs

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9. Ten Domain Expert Questions

These are questions a 43-year nuclear veteran would ask that Wall Street analysts wouldn't think of. Use these to evaluate these companies with your unique expertise.

1. Comanche Peak License Renewal Readiness

"What's the status of the Comanche Peak aging management program for the reactor vessel, containment tendons, and underground piping? Are there any known degradation issues that could complicate license renewal?"

Why it matters: License renewal (beyond 60 years to 80 years under Subsequent License Renewal) is not automatic. The NRC requires comprehensive aging management reviews. You've seen 43 years of the plant's history. You know where the bodies are buried — literally, what components have degraded, what repairs were deferred, what inspection findings were dispositioned. This is the kind of insight that determines whether VST's license renewal is a rubber stamp or a fight.

2. NuScale's Natural Circulation Claims

"NuScale claims their SMR uses natural circulation to eliminate coolant pumps and associated loss-of-coolant scenarios. Based on your understanding of PWR thermal-hydraulics, at what power level and containment configuration does natural circulation become unreliable? Have they adequately addressed boron dilution transients?"

Why it matters: Natural circulation is elegant but has physical limits. In large PWRs, forced circulation is necessary because natural circulation can't move enough coolant at full power. NuScale's 77 MWe size is small enough that natural circulation should work — but the NRC review of their safety analysis would reveal whether all transient scenarios (including boron dilution, which causes reactivity insertion) were adequately analyzed.

3. OKLO's NRC Denial — The Real Problem

"OKLO's 2022 combined license application was denied for insufficient safety analysis. As someone who's been through NRC licensing reviews, what does a denial at that stage tell you about the company's nuclear engineering depth? Is the deficiency fixable, or does it suggest fundamental gaps?"

Why it matters: Getting denied by the NRC isn't like failing a test you can retake. It means the staff reviewed your application and found it so deficient that they couldn't even begin the detailed technical review. This suggests either the company didn't have enough experienced nuclear engineers, or they underestimated what the NRC requires. Both are concerning for a company promising commercial reactors by 2030.

4. TRISO Fuel Qualification

"BWXT is producing TRISO fuel for Kairos and potentially others. Based on your understanding of nuclear fuel qualification, how confident are you that TRISO fuel performance data is sufficient for NRC licensing of commercial reactors? What's the biggest unknown?"

Why it matters: TRISO fuel has excellent test data from DOE programs, but commercial reactor licensing requires fuel performance data under all postulated accident conditions. The NRC will want to see fuel failure fractions under every scenario in the safety analysis. Curtis would know from I&C experience how instrumentation and control systems interact with fuel performance monitoring — a dimension Wall Street completely ignores.

5. Kairos Power's FLiBe Coolant Challenges

"Kairos uses fluoride salt (FLiBe) as a coolant with TRISO fuel. What are the real-world challenges of maintaining FLiBe chemistry, managing tritium production from lithium-6, and handling the corrosion issues that plagued the original MSRE at Oak Ridge? Are these engineering problems or science problems?"

Why it matters: The original MSRE ran successfully but revealed significant materials challenges — Hastelloy-N corrosion, tellurium embrittlement, tritium management. If these are engineering problems (we know the physics, just need better materials/processes), they're solvable on a commercial timeline. If they're science problems (we don't fully understand the mechanisms), Kairos is further away than their timeline suggests.

6. Emergency Planning Zone Implications for Data Center Co-Location

"If data centers are being built within 10 miles of nuclear plants (Comanche Circle near Comanche Peak), how does this affect the Emergency Planning Zone requirements? Does the NRC require emergency planning provisions for non-residential facilities in the EPZ?"

Why it matters: The EPZ is a 10-mile radius around every nuclear plant. Putting data centers inside the EPZ creates an interesting regulatory question: data centers have minimal staff but are critical infrastructure. Emergency evacuation of a data center isn't about people — it's about controlled shutdown of IT equipment. Curtis would understand the EP implications that real estate developers and tech companies haven't thought through.

7. Control Room Staffing for SMRs

"Current NRC regulations require licensed reactor operators in the control room 24/7. For multi-module SMR plants (NuScale proposes 4–12 modules per site), does each module need its own control room and operators, or can one control room oversee multiple modules? What are the I&C implications?"

Why it matters: Staffing is one of the biggest operating cost drivers for nuclear plants. If NuScale or OKLO can get NRC approval for reduced staffing (one operator overseeing multiple modules), the economics change dramatically. But from Curtis's I&C experience, he knows that control room design, alarm management, and human factors engineering for multi-module oversight is enormously complex. The NRC won't just wave through reduced staffing exemptions.

8. Supply Chain Reality for New Nuclear

"Based on your experience with ASME Section III vendors and nuclear-grade component procurement at Comanche Peak, how realistic is it to build 10+ SMRs per year by 2030? Where are the real bottlenecks — forgings, valves, instrumentation, QA-qualified welders?"

Why it matters: Everyone talks about building SMRs at scale. But Curtis has lived the supply chain reality. Nuclear-grade valves have 2–3 year lead times. ASME Section III welders are scarce. QA programs for nuclear suppliers take years to establish. The "factory-built modular" narrative assumes a supply chain that doesn't exist yet. Curtis can evaluate whether the timeline is realistic or fantasy.

9. Digital I&C Licensing — The Hidden Bottleneck

"The NRC has historically been skeptical of digital I&C systems in safety-related applications (software common-cause failure concerns). Advanced reactors all plan digital control systems. Based on your I&C experience, how big a licensing obstacle is digital I&C for SMR/OKLO/Kairos?"

Why it matters: Curtis spent years in I&C at Comanche Peak. He knows that the NRC's position on digital I&C safety systems has been one of the most contentious issues in nuclear licensing for decades. The NRC requires demonstration that software-based safety systems won't experience common-cause failure (a single software bug taking out all redundant safety channels). Every advanced reactor uses digital I&C. This is a hidden timeline risk that no Wall Street analyst understands.

10. Decommissioning Fund Adequacy

"For existing fleet operators (CEG, VST), are the decommissioning trust funds adequately funded given current cost estimates? Could underfunded decommissioning obligations become a balance sheet risk?"

Why it matters: Every nuclear plant must maintain a decommissioning fund (NRC requires it under 10 CFR 50.75). If the fund is underfunded, the operator must make additional contributions — a direct hit to cash flow. Conversely, if plants operate longer (80-year licenses), the fund has more time to grow. Curtis would know from Comanche Peak's financials whether the decommissioning funds are healthy, which informs the broader question of whether CEG and VST's nuclear assets are fully valued.

Appendix: Technical Data

Current Prices & Key Metrics (February 16, 2026)

Ticker	Price	Mkt Cap	P/E	Fwd P/E	Rev (TTM)	FCF	Beta	Short %
CEG	$288.43	$105B	33.0	25.1	$24.8B	$3.1B	1.14	—
VST	$171.49	$58B	61.7	18.7	$17.2B	$1.8B	1.44	—
GEV	$802.13	$218B	45.2	35.6	$38.1B	$5.3B	N/A	—
CCJ	$112.90	$49B	114.0	59.7	$3.5B	$348M	N/A	—
BWXT	$200.40	$18B	59.8	46.6	$3.1B	$273M	0.83	—
LEU	$199.19	$3.9B	51.1	37.9	$449M	$38M	1.21	25.0%
OKLO	$65.69	$10.3B	N/A	N/A	$0	-$41M	0.77	13.8%
SMR	$14.31	$4.0B	N/A	N/A	$64M	—	2.16	23.5%
NNE	$24.79	$1.3B	N/A	N/A	$0	-$25M	N/A	30.0%
FLR	$45.48	$7.4B	2.3	19.5	$15.6B	$299M	1.38	—
URA	$51.91	ETF	—	—	—	—	—	—

Uranium Market Snapshot

- **Spot Price:** ~$101/lb U_3O_8 (February 2026)
- **January 2026 Rally:** +24% in one month, back to triple digits
- **Structural Deficit:** Demand rising 28% by 2030, doubling by 2040
- **Key Producers:** Kazatomprom (Kazakhstan), Cameco (Canada), Orano (France)
- **Russian Enrichment:** ~40% of global SWU capacity — geopolitical risk factor
- **HALEU:** <20% enriched, required by most advanced reactor designs. Only Centrus (LEU) produces it domestically.

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